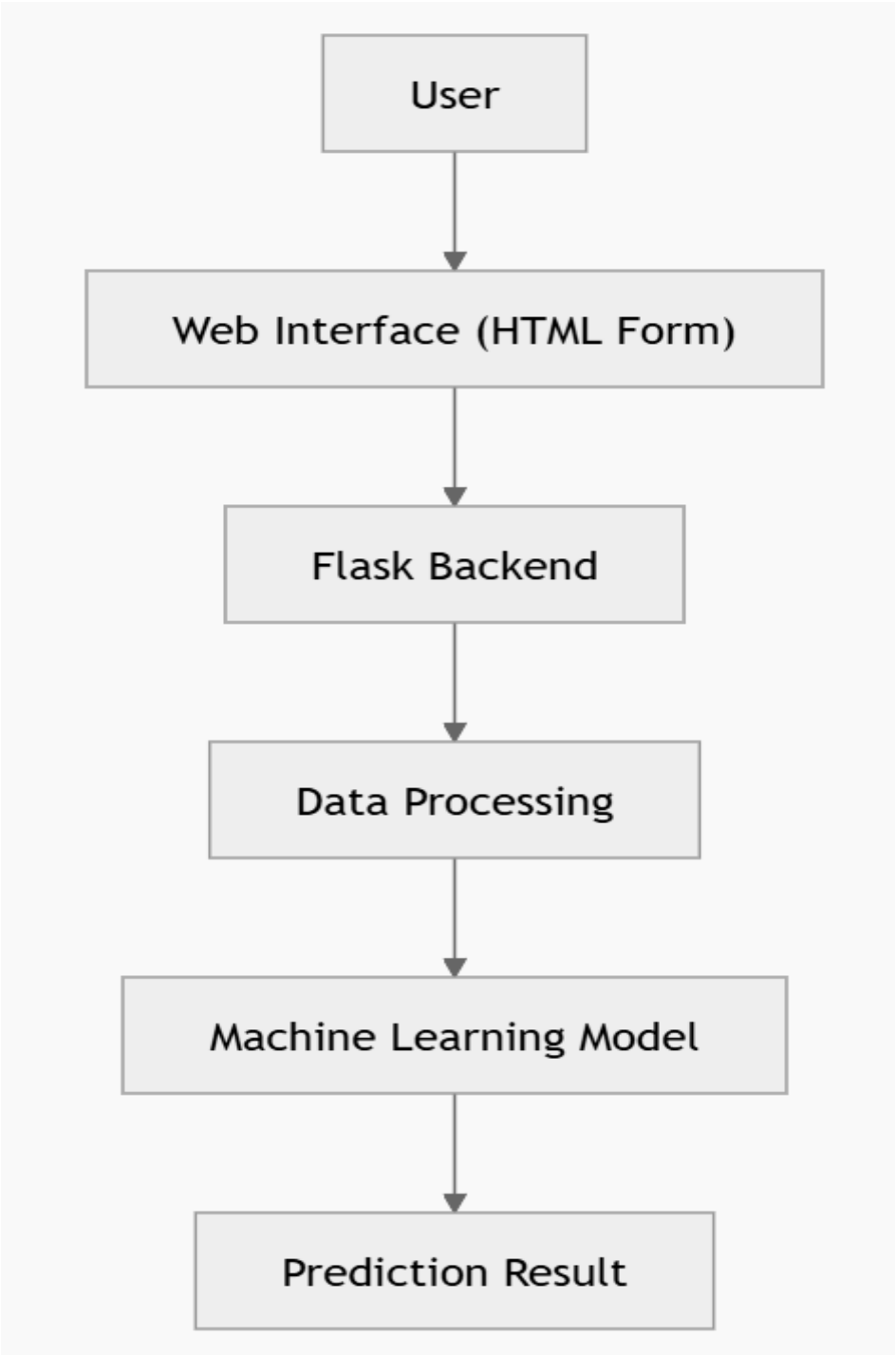
Project Design Phase II
Technology Stack (Architecture & Stack)

Date	8 March 2026
Team ID	699d5fa0c40b5ad0e9ee5a7d
Project Name	Predictive Pulse: Harnessing Machine Learning for Blood Pressure Analysis
Maximum Marks	4 Marks

Architecture Diagram



Components & Technologies

S.No	Component	Description	Technology
1	User Interface	Interface where users enter patient health parameters	HTML, CSS
2	Application Logic	Handles request processing and communication between UI and ML model	Python, Flask
3	Data Processing	Preprocesses and standardizes input data	Pandas, NumPy
4	Machine Learning Model	Predicts hypertension stage based on patient data	Scikit-learn
5	Model Storage	Stores trained ML model for prediction	Pickle (.pkl file)
6	Dataset	Patient health dataset used for training	CSV file
7	File Storage	Stores dataset and trained model files locally	Local File System
8	Infrastructure	Runs the Flask web application	Local System / Python Environment

Application Characteristics

S.No	Characteristics	Description	Technology
1	Open-Source Frameworks	Frameworks used for application development	Flask, Scikit-learn
2	Security Implementations	Input validation to prevent invalid data	Python validation logic
3	Scalable Architecture	Modular architecture separating UI, backend, and ML model	Flask-based architecture
4	Availability	Application accessible through web browser	Local server deployment
5	Performance	Fast prediction generation using trained ML model	Python ML libraries